Data Visualization
(B9DA106)

Tableau Practical Assignment

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LANDSLIDE EVENTS AROUND THE WORLD

NASA's EOSDIS maintains a comprehensive database called the Global Landslide Catalogue (GLC). It compiles data about landslides from numerous sources throughout the world. The catalogue contains information on landslides occurrences, such as their dates, locations, triggers, kinds, sizes, impacts, and casualties. The GLC is used by researchers and professionals to examine landslides trends, estimate hazards, and design mitigation solutions. It's a great resource for learning about landslides and building community resilience.

VARIABLE INFORMATION

- source_name: The name or source of the landslide event information.
- source_link: A link or reference to the source of the landslide event information.
- event_id: A unique identifier assigned to each landslide event in the catalogue.
- event_date: The date when the landslide event occurred.
- event_title: A title or brief description of the landslide event.
- event_description: A detailed description or summary of the landslide event.
- location_description: A description of the location or area where the landslide occurred.
- location_accuracy: An indicator of the accuracy or precision of the reported landslide location.
- landslide_category: The category or type of landslide, which could include categories such as rockslide, debris flow, or landslide complex.
- landslide_trigger: The triggering mechanism or cause of the landslide, such as rainfall, earthquake, or human activity.
- landslide_size: Information about the size or magnitude of the landslide event.
- landslide_setting: The environmental or geological setting in which the landslide occurred.
- fatality_count: The number of fatalities resulting from the landslide event.
- event_import_source: The source from which the landslide event information was imported or obtained.
- event_import_id: An identifier associated with the imported landslide event information.
- country_name: The name of the country where the landslide event occurred.
- country_code: A code representing the country where the landslide event occurred.
- admin_division_name: The name of the administrative division (e.g., state, province) within the country.
- admin_division_population: The population of the administrative division where the landslide occurred.
- gazeteer_closest_point: The closest known point of reference (e.g., town, landmark) to the landslide location.
- gazeteer_distance: The distance from the landslide location to the closest known point of reference.

- submitted_date: The date when the landslide event information was submitted to the catalog.
- created_date: The date when the entry for the landslide event was created in the catalog.
- last_edited_date: The date when the entry for the landslide event was last edited in the catalog.
- longitude: The geographical longitude coordinate of the landslide location.
- latitude: The geographical latitude coordinate of the landslide location.
- event_year: The year of the landslide event, extracted from the event_date.
- event_month: The month of the landslide event, extracted from the event_date.

OBJECTIVES OF VISUALIZATIONS

1. Fatality Rate by Landslide Trigger/Causes:

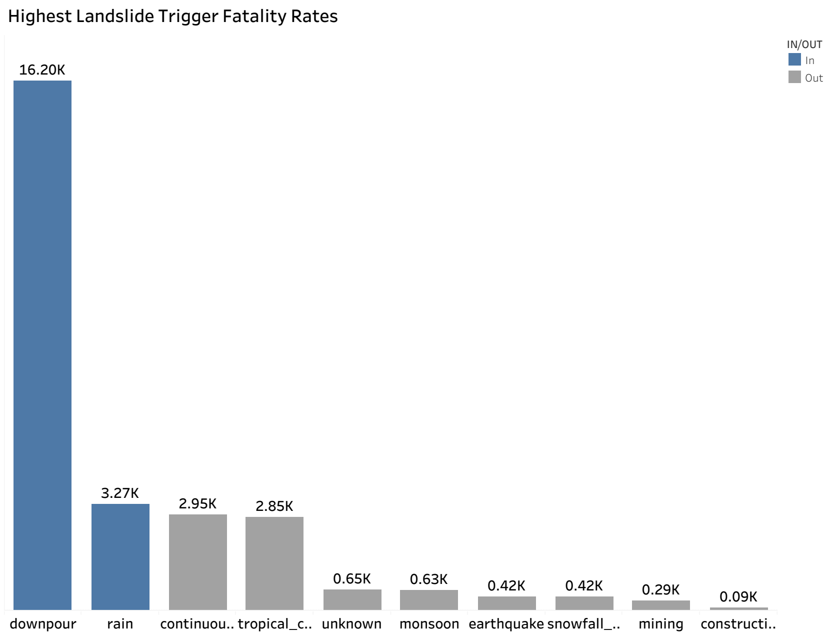

Fig 1 Fatality Rate By Landslide Trigger

From the graph which tries to offer insights into the fatality rate connected to various landslides triggers or causes. The goal of this visualisation is to investigate the link between the scale of landslides and the mortality rate in relation to different trigger types (such as rainfall, earthquakes, or human activities). By identifying the crucial regions for mitigation efforts and focused interventions, it helps identify the triggers and sizes that have the highest incidence of fatalities from the graph we can clearly see the most fatalities have occurred due to downpour and rain.

2. Fatalities Count by Countries:

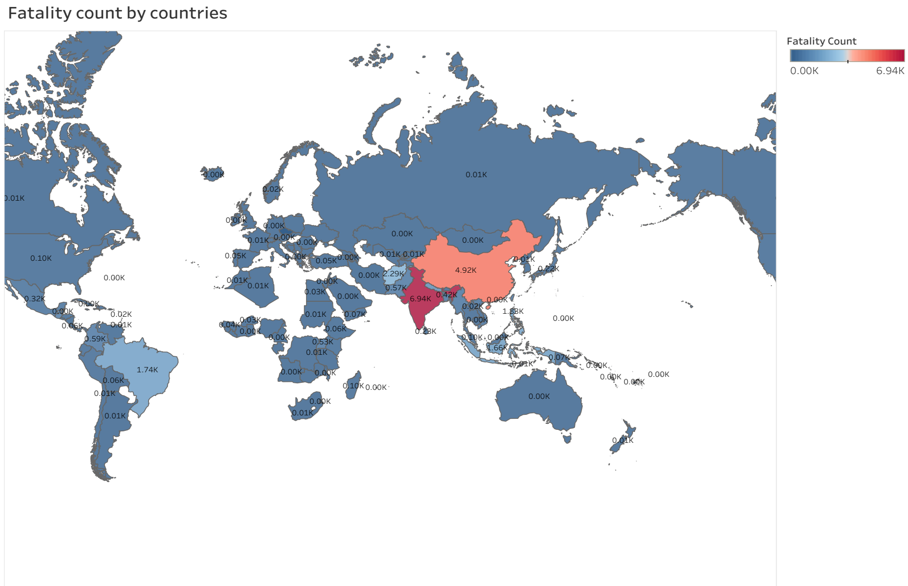

Fig 2 Fatalities Count By Countries

The number and amount of landslide deaths by nation are shown on this map visualisation. I created this map to give a general overview of the number of landslides-related fatalities in various nations. It is possible to identify places with greater risk and to prioritise resources and solutions by visualising the spatial distribution of casualties. It helps national or regional decision-making processes for risk reduction, preparedness, and disaster response plans.

3.Year-wise Fatalities:

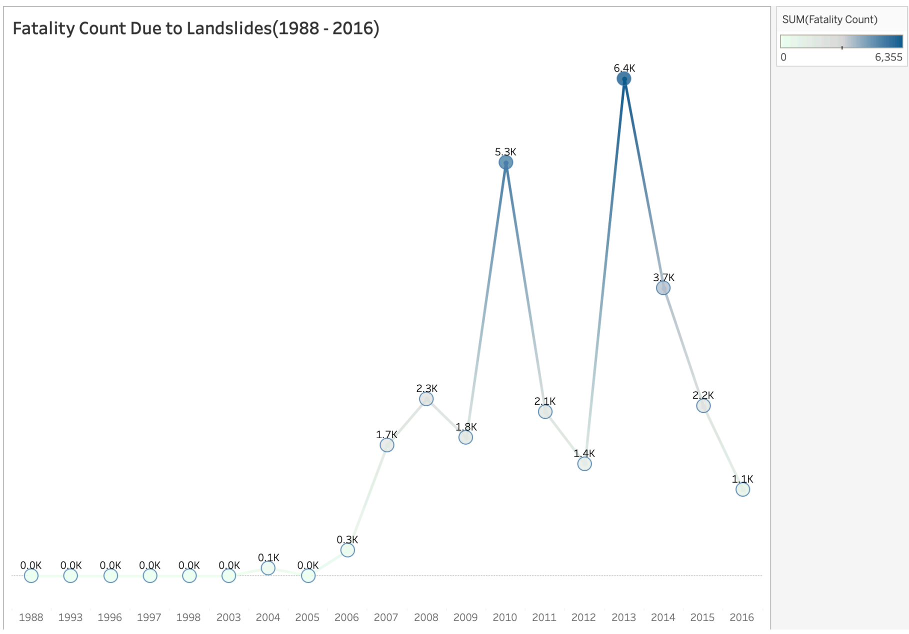

Fig 3 Year-Wise Fatalities

The goal of this line graph is to show how the number of deadly landslides that has changed through time.
I used this to show the temporal trends and patterns in landslides deaths throughout time. It is possible to identify times or particular years with notable shifts, spikes, or diminishing patterns by looking at the variations in fatalities. This data is essential for monitoring the reduction in landslide-related deaths as well as for evaluating the efficacy of policies and mitigating measures.

4. Number of Landslides by Countries:

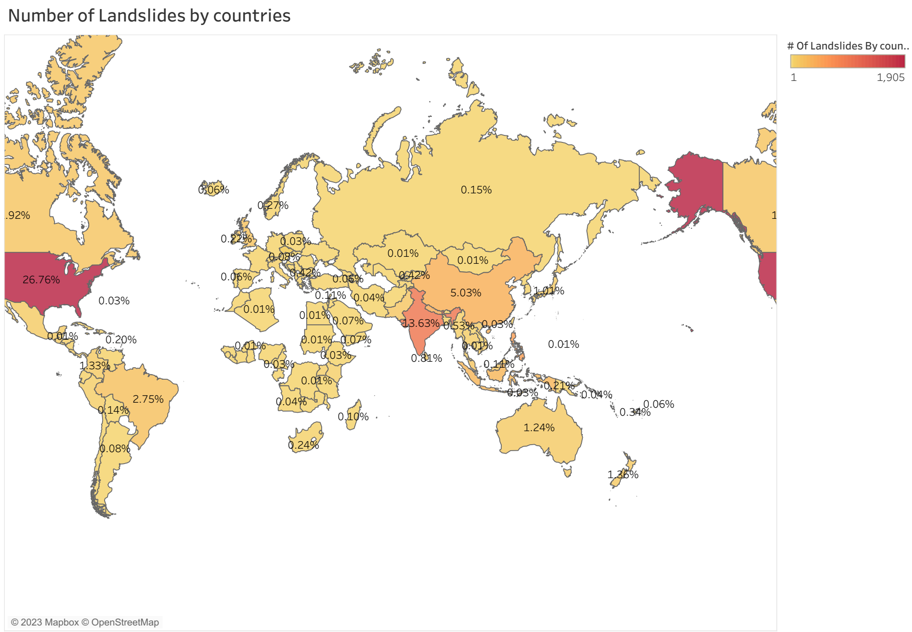

Fig 4 Number of landslides by countries

I made this map visualisation to show the distribution and % frequency of landslides that have happened in each nation.
By visualising the geographical distribution of landslides, it is possible to identify places with a greater occurrence of landslides, suggesting areas of potential high risk. From this map visualisation, we can observe the number of landslide incidents in various nations. This data helps with the allocation of resources for landslide management, prevention, and early warning systems as well as the prioritisation of research and monitoring activities.

INSIGHTS THAT I HAVE GAINED FROM THESE VISUALIZATIONS

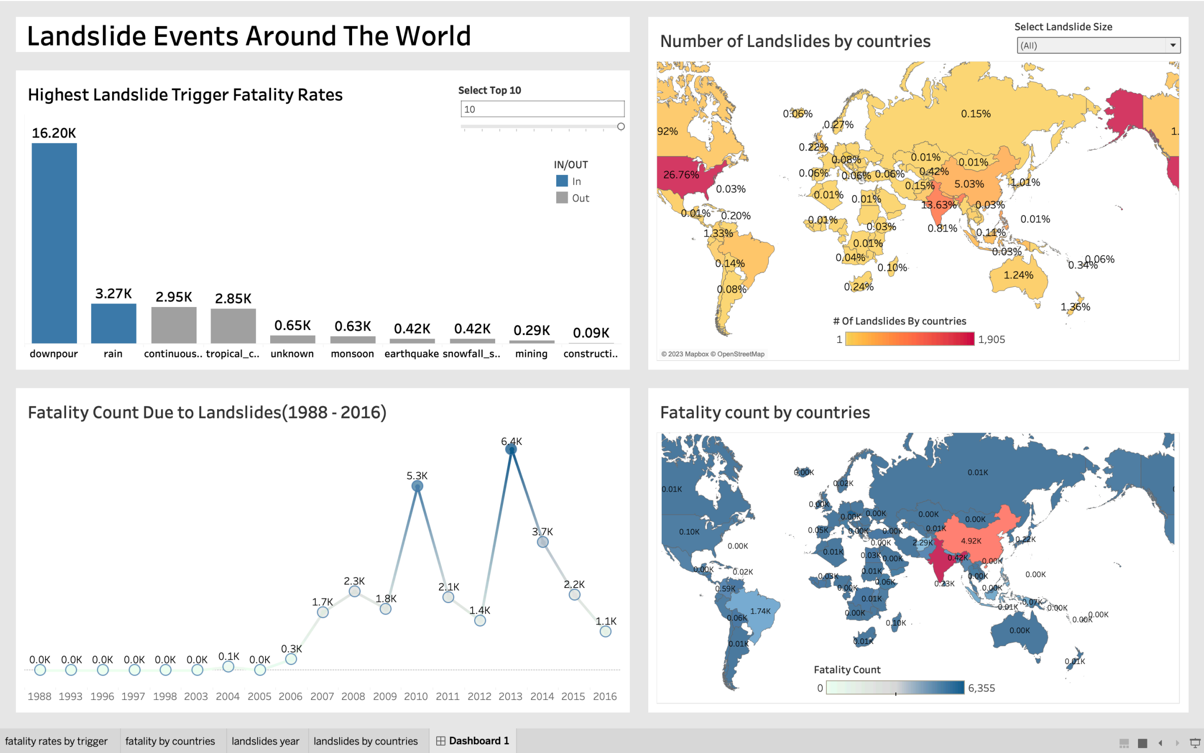

Fig 5 Landslide events around the world Dashboard

The first bar graph makes it very evident that certain causes and landslide magnitude have a big influence on mortality rates. It demonstrates that heavy rain and downpours are the main contributors to an increase in mortality following earthquakes, and that ongoing rains as well as human activities like mining and building are important causes linked to fatality rates. This realisation emphasises how crucial it is to concentrate on these triggers while putting mitigation strategies and disaster response plans in place. We can create targeted measures that decrease the effects of landslides by analysing the relationship between trigger types, landslide magnitude, and mortality rates. We can also observe that compared to other nations, India, China, Brazil, and Afghanistan have a greater rate of mortality. These results highlight the necessity for these nations to give catastrophe risk reduction and management top priority. Years like 2013 and 2014 saw very high rates of fatalities from landslides. This realisation inspires more research into the elements and occasions unique to those years. Extreme weather, geological conditions, or other reasons that contributed to higher death rates could be to blame. Understanding the factors that led to these years' increased mortality rates can help develop focused treatments and risk-reduction measures. Other nations like the USA has very larger amount of landslides than China, India, and Brazil. It implies that these areas require efficient landslide monitoring, land-use planning, and risk mitigation strategies. Individuals or organizations in the that sector can prioritise resources and strategies to minimise the incidence and possible repercussions of landslides by studying the concentration and spatial distribution of landslides. In order to reduce the effects of landslides during times of high susceptibility, it emphasises the significance of thorough risk assessments, effective early warning systems, and disaster management practises.

The results of these visualizations help me understand the relationships between landslide causes, mortality rates, regional distribution, and temporal fluctuations. They provide helpful suggestions for creating focused strategies and actions to organizations and professionals in disaster management. It is feasible to decrease the consequences of landslides, improve community resilience, and save lives by addressing the particular triggers, high-fatality zones, and temporal patterns.